Semester Project Documentation (DSA – CS – 221)

Semester Project Title:

Raju Food Ordering and Delivery Management System

Name	**Registration No.**	**Degree**
M. Daniyal	2024373	EE
Muzammil Faisal	2024687	EE
Zohaib Ali	2024687	EE

1. **Main Features**
    1. User Registration and Login System for Students, Admin, Kitchen Staff, and Delivery Users.
    2. Menu Management System - Allows to add, remove, search, filter, and sort food items
    3. Cart Management System - Enables students to add, remove, update items in cart
    4. Order Placement and Tracking - Enables students to place orders and track their current status from placement until delivery.
    5. Order Processing Status - Manages order lifecycle (Pending, Preparing, Ready, Delivering, Delivered)
    6. Delivery Management - Assigns and delivers ready orders.
    7. Graph-Based Hostel Routing System - Represents hostel locations and distances in graph form for delivery visualization
    8. File Handling - Stores and retrieves users, menu, and orders using text files
2. **Types of Users & Requirements**
    1. **Student (Customer):**

    1.1. Will be able to register using reg number and phone number.

    1.2. Will be able to login using reg numbers and password

    1.3. Will be able to view menu

    1.4. Will be able to sort menu

    1.5. Will be able to manage cart (Add Items, Remove, Update quantity)

    1.6. Will be able to undo last cart action

    1.7. Will be able to place orders and track order status.

    2. **Kitchen Staff:**

    2.1. Will be able to login by given username and password from admin.

    2.2. Will be able to view and update status for pending orders.

    2.3. Will be able manage menu items (View, add new item, remove, Update price, toggle availability)

    3. **Delivery Staff:**

    3.1. Will be able to login via username and password set by Admin.

    3.2. Will be able to view ready orders

3.3. Will be able to accept deliveries to hostels

3.4. Will be able to see their active deliveries

3.5. Will be able to mark delivered once delivered are delivered.

3.6. Will be able to see Campus Map to see the route and distance.

4. Admin:

4.1. Will be able to login using admin username and password.

4.2. Will be able to view all users (students, Delivery users, Kitchen Users)

4.3. Will be able to create new staff accounts.

4.4. Will be able to view all orders

4.5. Will be able to view Campus map

3. Requirements Breakdown

1. User Registration and Login System

1.1. The system shall display login and registration options to users at startup.

1.2. The system shall allow students to register using registration number, phone number, and password.

1.3. The system shall validate registration number and phone number format during registration.

1.4. The system shall store registered user data in text files for persistence.

1.5. The system shall authenticate users by matching login credentials with stored file data.

1.6. The system shall support role-based login for Admin, Kitchen Staff, and Delivery Staff.

2. Menu Management System

2.1. The system shall allow viewing of all available food items from the menu.

2.2. The system shall allow menu items to be added by authorized staff.

2.3. The system shall allow menu items to be removed from the menu by authorized staff.

2.4. The system shall allow updating item price and availability status.

2.5. The system shall support searching menu items.

2.6. The system shall support filtering menu items by category.

2.7.The system shall allow sorting menu items by price and name.

3. Cart Management System

3.1.The system shall allow students to add items to the cart.

3.2.The system shall allow students to remove items from the cart.

3.3.The system shall allow students to update item quantities in the cart.

3.4.The system shall calculate total cart price dynamically.

3.5.The system shall allow undoing the last cart operation.

4. Order Placement and Tracking System

4.1.The system shall allow students to place orders from cart items.

4.2.The system shall generate a unique order ID for each order.

4.3.The system shall store order details including items, price, hostel, and room.

4.4.The system shall allow students to view their placed orders.

4.5.The system shall display real-time order status to students.

5. Order Processing Status Management

5.1.The system shall assign initial status *Pending* to newly placed orders.

5.2.The system shall allow kitchen staff to mark orders as *Preparing*.

5.3.The system shall allow orders to be marked *Ready* after preparation.

5.4.The system shall allow delivery staff to update order status to *Delivering/Delivered*.

6. Delivery Management System

6.1.The system shall enqueue ready orders into the delivery queue.

6.2.The system shall allow delivery staff to view available delivery orders.

6.3.The system shall remove completed deliveries from the delivery queue.

7. Graph-Based Hostel Routing System

7.1.The system shall represent hostel locations and distances using a graph structure.

7.2.The system shall display hostel distance information for delivery visualization.

7.3.The system shall allow delivery staff to view campus map and distance details.

8. File Handling and Data Persistence

8.1.The system shall store users, menu items, and orders in text files.

8.2.The system shall load stored data at program startup.

8.3.The system shall update files after any add, remove, or update operation.

8.4.The system shall preserve order history across multiple executions.

4. Feature to Coding Matrix

Sr.	Feature	DSA Concept	Operations	Complexity	No. of Variables & Objects	Functions	Lines of Code
1.	User Login	Hash maps	Search, Insert	O(1)	6	11	259
2.	Menu System	Array	Traversal, Linear Search, Insertion, Deletion	O(n) O(1) - Insert	5	10	255
3.	Menu Sorting	Bubble Sort Algorithm	Compare and Swap	$O(n^2)$	Same as Menu System	2	255
4.	Menu Search	Linear Search	Substring Search	O(n)	Same as Menu System	1	255
5.	Cart System	Stack + Array	Push/Pop (Undo), Add/Remove Items	O(1)	5	9	200

6.	Orders System	Singly Linkedlist	Insert, Delete, Search, Traversal	O(n)	8	11	688
7.	Kitchen and Delivery System	Queue (circular)	Enqueue/Dequeue	O(1)	5	7	688
8.	Campus Map	Graph Adjacency Matrix	Add Vertex/Edge, Search, Displa	**O(1)** edge lookup, **O(n)** search	4	5	153

5. Project Screenshots

Front Screen

```
========================================
   RAJU Food Ordering System - GIKI
========================================

Press Enter to continue...
```

Registration/Login Screen

```
========== RAJU FOOD SYSTEM - MAIN MENU ==========
1. Register (Students Only)
2. Login
3. Exit
==================================================

Note: Staff credentials stored in data/admin_users.txt
Enter your choice:
```

Student Registration

```
========== STUDENT REGISTRATION ==========
Note: Only students can register here.
Staff accounts managed by administration.
==========================================

Enter Full Name: Daniyal
Enter Registration Number (7 digits, e.g., 2024001): 2024373
Enter Phone Number (11 digits, e.g., 03001234567): 03089435928
Enter Password: 123

Registration Successful!

--- User Information ---
Name: Daniyal
Phone: 03089435928
Reg Number: 2024373
Role: Student

You can login using your Registration Number: 2024373

Press Enter to continue...
```

Login Screen

```
========== USER LOGIN ==========
1. Student Login
2. Staff Login
3. Back
================================
Enter choice:
```

Student Screen – after Login

```
========== STUDENT MENU ==========
Welcome, Daniyal!
Reg #: 2024373
Cart Items: 0 | Total: Rs.0
===================================
1. View Full Menu
2. Browse by Category
3. Search Menu
4. My Cart
5. Place Order
6. Order History
7. Logout
===================================
Enter your choice:
```

Menu Display for Students

```
========== RAJU FOOD MENU ==========
ID   Item Name                Price       Category      Status
---------------------------------------------------------------------
1    Chicken Burger           Rs.  1111   meal         Available
3    Beef Burger              Rs.   280   meal         Available
4    Fries (Regular)          Rs.   100   snack        Available
5    Fries (Large)            Rs.   150   snack        Available
6    Samosa                   Rs.    20   snack        Available
7    Roll                     Rs.    80   snack        Available
8    Cold Drink               Rs.    50   drink        Available
9    Tea                      Rs.    30   drink        Available
10   Coffee                   Rs.    60   drink        Available
11   abc                      Rs.   100   meal         Available
12   aa                       Rs.     2   meal         Available
=====================================================================

Options:
1. Sort by Price (Low to High)
2. Sort by Name (A-Z)
3. Back to Main Menu
Enter choice:
```

Cart Options for Students

```
Your cart is empty.

========== CART OPTIONS ==========
1. Add Item to Cart
2. Remove Item from Cart
3. Update Quantity
4. Undo Last Action
5. Clear Cart
6. Proceed to Checkout
7. Back to Menu
==================================
Enter choice:
```

Adding item to cart - Student

```
========== RAJU FOOD MENU ==========
ID   Item Name                Price       Category      Status
----------------------------------------------------------------------
1    Chicken Burger           Rs.  1111   meal        Available
3    Beef Burger              Rs.   280   meal        Available
4    Fries (Regular)          Rs.   100   snack       Available
5    Fries (Large)            Rs.   150   snack       Available
6    Samosa                   Rs.    20   snack       Available
7    Roll                     Rs.    80   snack       Available
8    Cold Drink               Rs.    50   drink       Available
9    Tea                      Rs.    30   drink       Available
10   Coffee                   Rs.    60   drink       Available
11   abc                      Rs.   100   meal        Available
12   aa                       Rs.     2   meal        Available
======================================================================

Enter Item ID (0 to cancel): 1
Enter Quantity: 1
Added Chicken Burger to cart.

Press Enter to continue...
```

Order Placement - Student

```
========== YOUR CART ==========
ID   Item                        Price     Qty    Subtotal
-------------------------------------------------------------
1    Chicken Burger        Rs.  1111       1 Rs.   1111
=============================================================
                                TOTAL: Rs. 1111
=============================================================

========== ORDER DETAILS ==========

Enter Hostel (e.g., H1, H2, GH1): H10
Enter Room Number: 11

========== ORDER SUMMARY ==========
Delivery to: H10, Room 11
Total: Rs.1111
===================================

Confirm order? (y/n): y

Order placed successfully! Order ID: 4
Your order has been sent to the kitchen.

Order #4 placed successfully!
You will be notified when it's ready.
Cart cleared.

Press Enter to continue...
```

Order History - Student

```
========== YOUR ORDERS ==========
ID    Name                Total       Status         Location
-----------------------------------------------------------
4     Daniyal             Rs.    1111 Pending       H10-11

Press Enter to continue...
```

Student Screen for Order History – After Delivery

```
========== YOUR ORDERS ==========
ID    Name                Total       Status         Location
-----------------------------------------------------------
4     Daniyal             Rs.    1111 Delivered     H10-11

Press Enter to continue...
```

Admin Login

```
========== STAFF LOGIN ==========
Enter Username: admin
Enter Password: admin

Login Successful!
Welcome, Admin User!

Press Enter to continue...
```

Admin Main Screen

```
========== ADMIN MENU ==========
Welcome, Admin User!
================================
1. View All Users
2. Create Staff Account
3. View All Orders
4. View Campus Map
5. Logout
================================
Enter your choice:
```

Kitchen Login

```
========== STAFF LOGIN ==========
Enter Username: kitchen
Enter Password: 123

Login Successful!
Welcome, kitchen!

Press Enter to continue...
```

Kitchen Staff Main Screen

```
========== KITCHEN STAFF MENU ==========
Welcome, kitchen!
========================================
1. View Pending Orders
2. Update Order Status
3. Manage Menu Items
4. Logout
========================================
Enter your choice:
```

Pending Order View

```
========== KITCHEN - PENDING ORDERS ==========
ID    Student          Total     Location       Time
-----------------------------------------------------------------
4     Daniyal          Rs.  1111 H10-11        22:23
-----------------------------------------------------------------
Total: 1 order(s)

========== KITCHEN - PREPARING ORDERS ==========
No orders with this status.

Press Enter to continue...
```

Update Order Status Screen

```
========== PENDING ORDERS (Not Started) ==========
ID    Student           Total     Location       Time
-----------------------------------------------------------------
4     Daniyal           Rs.  1111 H10-11        22:23
-----------------------------------------------------------------
Total: 1 order(s)

========== PREPARING ORDERS (In Kitchen) ==========
No orders with this status.

========== UPDATE ORDER STATUS ==========
1. Start Preparing Next Order (from queue)
2. Mark Order as Ready (move to delivery)
3. Back
==========================================
Enter choice:
```

Kitchen Staff managing Menu Items

```
========== MENU MANAGEMENT ==========
1. View All Items
2. Add New Item
3. Remove Item
4. Update Item Price
5. Toggle Availability
6. Back
=====================================
Enter choice:
```

Delivery Staff Login

```
========== STAFF LOGIN ==========
Enter Username: delivery
Enter Password: delivery

Login Successful!
Welcome, Delivery Boy!

Press Enter to continue...
```

Main Screen for Delivery Staff

```
========== DELIVERY PERSONNEL MENU ==========
Welcome, Delivery Boy!
=============================================
1. View Ready Orders
2. Accept Delivery
3. My Active Deliveries
4. Mark as Delivered
5. View Campus Map
6. Logout
=============================================
Enter your choice:
```

Accept Deliveries via Order ID

```
========== READY FOR DELIVERY ==========
ID    Student          Total     Location       Time
-------------------------------------------------------
4     Daniyal          Rs.  1111 H10-11        22:23
-------------------------------------------------------
Total: 1 order(s)

========== DELIVERY QUEUE ==========
Orders ready: 1
Next order: #4
====================================

========== ORDER DETAILS ==========
Order ID: 4
Student: Daniyal (03089435928)
Location: H10, Room 11
Status: Ready
Order Time: 2025-12-21 22:23:21

Items:
----------------------------------
Chicken Burger      x1 = Rs.1111
----------------------------------
Total: Rs.1111
==================================

Accept next delivery? (y/n): y
Enter Order ID to accept: 4
Order #4 assigned to Delivery Boy

Press Enter to continue...
```

Mark Orders as Delivered

```
========== ORDERS OUT FOR DELIVERY ==========
ID    Student          Total     Location       Time
-----------------------------------------------------------------
4     Daniyal          Rs.  1111 H10-11       22:23
-----------------------------------------------------------------
Total: 1 order(s)

Enter Order ID to mark as delivered (0 to cancel): 4
Order #4 marked as delivered!

Ôô£ô Order #4 marked as delivered!
Payment collected. Thank you!

Press Enter to continue...
```

Campus Graph

```
==========================================================================
                    GIKI CAMPUS MAP - RAJU HOTEL
==========================================================================

                        Main Gate (300m)
                             |
                             |
    Medical Center -------- Raju Hotel (Kitchen)
         |  (250m)           |    |    |
         |                   |    |    |
    Girls H7 (200m)      150m 200m 250m
         |                   |    |    |
         |                   v    v    v
    Girls H13              H1--H2--H3--H4--H5--H6
      (150m)                 |                |
                             |              H8--H9--H10--H11--H12
                        Main Gate

--------------------------------------------------------------------------
DELIVERY LOCATIONS:
--------------------------------------------------------------------------
Location                 Type
--------------------------------------------------------------------------
Raju Hotel               Kitchen/Pickup Point
Boys Hostels H1-H6       Delivery Zone A
Boys Hostels H8-H12      Delivery Zone B
Girls Hostels H7, H13    Delivery Zone C
Main Gate                Campus Entry
Medical Center           Medical Facility
--------------------------------------------------------------------------

TOTAL LOCATIONS: 16
==========================================================================

Press Enter to continue...
```